

- 6 (a) Frequency is the number of complete oscillations per unit time, measured in hertz (Hz).

Angular frequency is the product of frequency and the angle subtended by the circumference of a circle (2π radians). It is a measure of the rate of change of angular displacement of the motion with time, measured in radians per second (rad s^{-1}).

Note: Very few gave $\omega = 2\pi f$, with explanation.

(b) (i) Loss in gravitational potential energy = $(0.400)(9.81)(0.200) = 0.785 \text{ J}$

(ii) $k = \frac{F}{e} = \frac{0.400(9.81)}{0.200} = 19.62 \text{ N m}^{-1}$

Elastic potential energy stored = $\frac{1}{2}(19.62)(0.200)^2 = 0.392 \text{ J}$

(c) The difference is work done against the external force needed to support the mass while lowering it gently. This force is the difference between the mass's weight and the tension in the spring. The work done is dissipated as heat.

Note: Very few seemed to realise that the load was lowered gently.

(d) (i) At the lowest point, the total extension of the spring is 0.400 m.

Tension $T = ke = (19.62)(0.400) = 7.848 \text{ N}$

Weight $W = mg = (0.400)(9.81) = 3.924 \text{ N}$

Resultant force = $T - mg = 7.848 - 3.924 = 3.92 \text{ N}$

(ii) Angular frequency of mass-spring system

$\omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{19.62}{0.400}} = 7.00 \text{ rad s}^{-1}$

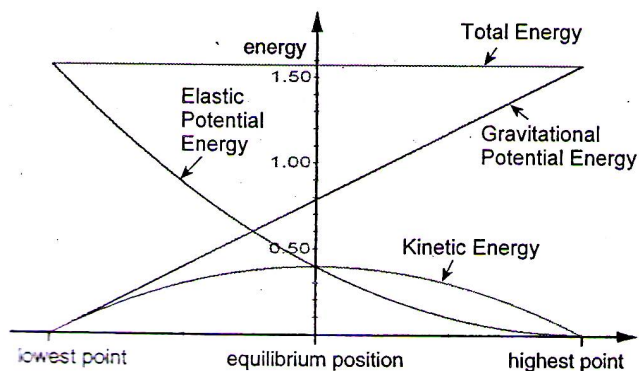
(iii) Maximum speed

$v_0 = \omega x_0 = (7.00)(0.200) = 1.40 \text{ m s}^{-1}$

(e)

	gravitational potential energy / J	elastic potential energy / J	kinetic energy / J	total energy / J
lowest point	0	1.57	0	1.57
equilibrium position	0.785	0.392	0.392	1.57
highest point	1.57	0	0	1.57

(f)



7 (a) The evidence is adduced from Rutherford's α -scattering experiment